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# Paleomagnetic results from volcanic rocks of the Shelve Inlier, Wales: evidence for a wide Late Ordovician Iapetus Ocean in Britain

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In order to shed light on the early Paleozoic paleogeography of Iapetus-bordering continents, samples for paleomagnetic study were collected from middle Ordovician (Llanvirn) volcanic rocks of the Shelve Inlier, Welsh Borderland. The folding within the inlier is constrained to late Ordovician (Ashgill) time, thus offering a fold test with tight age brackets. After stepwise thermal demagnetization, a pre-folding characteristic magnetization component was isolated from 95 specimens. The mean of 11 site mean directions for the characteristic magnetization gives $D/I = 116.4/+67.9$ ($k = 89.4$; $\alpha_{95} = 4.9°$), and the paleomagnetic pole lies at 26.6° N, 36.1° E ($\delta p = 6.9°$, $\delta m = 8.2°$). The fold test is positive at the 99% confidence level, indicating that the magnetization was acquired during the middle or late Ordovician. We infer from geological considerations that this magnetization is not a primary thermoremanence, but was acquired during late Ordovician time as a result of a thermochemical disturbance of that age. Our results indicate that southern Britain was near 51° south latitude during the later Ordovician. Since middle to late Ordovician paleolatitudes for Gondwana and Laurentia are well-determined, our result can be used to test various scenarios for the closure of Iapetus. We conclude that: (1) the Iapetus Ocean in Britain still had a latitudinal width of some 30° at this time, thus rendering a postulated Ordovician closure unlikely; (2) southern Britain, Avalon, and Hercynian Europe all occupied high latitudes during the later Ordovician, consistent with a paleoposition near the northern margin of Gondwana; and (3) the Acadian orogeny in Britain and North America could have been the result of the collision of Gondwana and Laurentia during early Devonian time.

## 1. Introduction

A radically revised picture of Paleozoic paleogeography is emerging from recent paleomagnetic studies in Atlantic-bordering continents which is broadly consistent with biogeographic and paleoclimatic indicators [1]. The North American apparent polar wander (APW) path for the Paleozoic has recently been revised by paleomagnetic analyses using detailed demagnetization procedures to isolate individual magnetization components, and fold tests to constrain the age of the magnetizations (e.g. [2–5]). Paleomagnetic data from Africa [6], Avalonian basement blocks in North America [7], and from the Armorican and Iberian Massifs [8,9] provide data which paint a paleogeographic picture for the Ordovician in which Avalon and Hercynian Europe were part of (or close to) the margin of Gondwana and were at high southerly latitudes, with the south pole being located in northern Africa. North America was situated in equatorial latitudes at this time, and a several thousand kilometer wide Ordovician Ocean separating Gondwana and Laurentia is indicated [1]. A new paleomagnetic pole for cratonic Africa [10] indicates that the northern margin of Africa had moved northward into the tropical zone by Silurian time, a conclusion which is consistent with mapping of middle Paleozoic glacial centers across Gondwana [11]. Although the timing of this major shift in the position of Gondwana is not yet entirely clear, it does appear that by Silurian–Devonian time the wide Ordovician ocean separating Laurentia and Gondwana could have been closed.

Southern Britain (i.e., south of the Iapetus suture) remains a puzzle-stone in this paleogeographic scheme. The suture of the Iapetus Ocean in Britain is usually placed to the south of the Southern Uplands Terrane (close to the England/Scotland border) passing southwestward through central Ireland. Commonly-cited Ordovician

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paleomagnetic poles from Britain south of the suture (e.g. [12]) indicate intermediate paleolatitudes (about 30°S), suggesting (in the scenario outlined above) a paleogeographic position intermediate between Gondwana and Laurentia. A relatively narrow Ordovician Iapetus Ocean between southern Britain and Laurentia has gained some support from one contingent in the "Southern Uplands controversy" [13]. Whereas the Southern Uplands is widely regarded as an accretionary complex developed during the Ordovician/Silurian subduction of Iapetus oceanic lithosphere [14,15], alternative hypotheses have been aired in the last few years (e.g. [16–19]). These authors advocate late Ordovician closure of the Iapetus Ocean in this area, precluding the possibility that the Silurian sedimentary sequence in the Southern Uplands can have been deposited as an accretionary complex. They cite sedimentological evidence for distinct external and internal detrital sources which imply that the Southern Uplands Basin was either a back-arc/foreland basin [18,19] or a "successor" basin [17] largely developed subsequent to closure of the Iapetus Ocean. However, these arguments have been criticized on the basis of findings in modern accretionary prisms, which indicate that the observed depositional features in the Southern Uplands are consistent with active-margin processes [20].

Evidence for late Ordovician–early Silurian homogenization of shelf faunas across the Iapetus suture [21,22] can be interpreted as favoring a narrow Iapetus Ocean at this time. In addition, it has been suggested on the basis of tectonostratigraphic correlations that the first continental contact was between southern Britain and western Newfoundland in Ashgill time [23], implying several thousand kilometers of post-collisional sinistral displacement along the suture between the late Ordovician and the early Devonian. On the other hand, the proposed Newfoundland sourcing of granitic pebbles in Caradoc rocks of the Southern Uplands [24] has been countered by the observation that Precambrian basement of similar lithology and yielding similar radiometric ages is exposed in northwestern Ireland, and therefore may have constituted the basement over a wider area [25].

The paleomagnetic data from Britain south of the Iapetus suture should provide the critical constraints. Paleozoic paleomagnetic data from Britain have been reviewed by Briden et al. [12] and Piper [26]. Ordovician paleolatitudes from Britain south of the suture are generally in the mid range (~ 30°S) with paleopoles of low latitude and longitude. However, a number of studies have yielded "anomalous" high inclination data yielding higher paleolatitudes. These data include results from volcanics and intrusives from North Wales [27], intrusives from the Welsh borderlands [28], and Lake District volcanics and intrusives [29]. Whereas Briden et al. [12] consider these data to be aberrant and do not include them in their apparent polar wander path, Piper [26] attaches importance to these paleopoles and considers that they link the low latitude Ordovician paleopoles with antipodal low latitude Silurian paleopoles. These contrasting interpretations of the early Paleozoic paleomagnetic data imply strongly contrasting paleogeographies. The significance of the high inclination data is difficult to evaluate. Most of these results are from intrusive units for which the paleohorizontal is unknown or poorly constrained. Moreover, the data were obtained without detailed demagnetization experiments or field tests to bracket the magnetization ages.

The Paleozoic paleogeography advocated by Van der Voo [1] implies rapid closure of a wide Ordovician Iapetus Ocean in the Silurian culminating in the early Devonian (Acadian) collision of Gondwana and Laurentia. In order to determine the role of southern Britain in this scenario, paleomagnetic data with tight control on the age of the magnetization components must be acquired. Previously published Ordovician paleomagnetic data from southern Britain, with the notable exception of the recent data from the Builth Inlier [30], have been acquired using blanket demagnetization, mostly with the alternating field method. As demonstrated by the data from the Builth Inlier, detailed stepwise thermal demagnetization is required, at least in this case, to resolve components in a complex multicomponent magnetization history. If individual magnetization components are not isolated by detailed demagnetization procedures, fold tests do not validate the resulting magnetization directions (e.g. [3]).

Positive fold tests from the Borrowdale Volcanics [29,31] and from the Shelve Inlier [28] suggest that a pre-folding component is present in the

magnetization directions, but it has not been demonstrated with stepwise demagnetization experiments that the magnetizations are single components. The paleomagnetic poles obtained in these investigations are therefore in need of revision using more modern techniques. In the case of the Builth study [30], a characteristic magnetization appears to have been successfully isolated during detailed laboratory study. A conglomerate test based on 6 clast directions from a syn-depositional volcanic agglomerate was offered to bracket the age of the characteristic magnetization. Although the conglomerate test was statistically positive, we note that of 5 of the 6 clast directions are not randomly distributed, and that their mean is close to the characteristic direction obtained in the main study. We therefore regard the test to be inconclusive, and the age of the Builth magnetization unconstrained.

Although previously published site mean magnetization directions from the Shelve Inlier volcanic rocks [28] pass the fold test suggesting an Ordovician magnetization age, the directions are significantly different from the steeper directions from the Ordovician plutonic rocks in the area [28]. The purpose of this paper is to re-examine the paleomagnetism of the Shelve extrusive rocks using thermal demagnetization techniques and modern component analysis to attempt to resolve individual magnetization components and apply the fold test to ascertain the relative age of the characteristic magnetization.

## 2. Geological setting and sampling

Our primary objective in this investigation has been to find an Ordovician magnetization for southern Britain with a well-determined acquisition age, constrained by a fold test. Folded Ordovician volcanics crop out in several areas, including the English Lake District, north and south Wales, and the Welsh Borderland region. Most of these areas are north of the Acadian Front (Fig. 1), and were subject to folding and cleavage development during early Devonian time [33,34]. A small area in Wales that includes the Builth and Shelve Ordovician Inliers lies between the Acadian and Hercynian fronts (Fig. 1). This area was subject to one principal phase of deformation, during the late Ordovician (Ashgill), and is therefore unique in offering the possibility of constraining a magnetization to Ordovician time. This study was conducted in the Shelve Inlier, which straddles the Wales/England border in Powys and Shropshire counties (Fig. 1).

The significance of the Ashgill deformation in the Builth and Shelve Inliers is controversial. It has been suggested that the northeast-trending folds and faults are a product of southeast-directed thrust tectonics related to the closure of the intervening Iapetus Ocean (e.g. [35]). However, more recently, late Ordovician deformation in this area has been associated with dextral strike-slip movement along the Pontesford Lineament and related northeast-oriented faults [36,37]. Some of these faults were active from at least mid-Ordovician to Triassic, but the most important movement phase was Ashgill–early Llandovery in age [36]. As the Upper Llandovery to Wenlock cover of the inliers is largely undeformed, and the youngest deformed rocks are Lower Caradoc in age, the deformation is locally bracketed to the interval between early Caradoc and late Llandovery. Elsewhere in the Welsh Basin, this deformation phase is more tightly constrained to Ashgill time [37].

Detailed studies of the Shelve Inlier [38] and the Builth Inlier [39] relate the Ashgill deformation to dextral shear in a complex zone of deformation. In the Shelve Inlier, none of the major faults is exposed and the evidence for their existence is based on detailed mapping. The details of the stratigraphy and structure of the Inlier are given in a series of eight Open File Reports of the British Geological Survey with accompanying 1 : 10,000 scale maps (see [38]). From cleavage and bedding orientations, it is known that folds within the inlier have vertical or sub-vertical axial planes and offset of axial planes is well documented [38]. The minimum possible slip necessary to explain the observed structural and lithological displacements suggests that the displacements are predominantly strike-slip. Displacements on individual strike-slip faults are generally less than 2 km, but the aggregate across the inlier must have been considerable. The deformation has been related to dextral transpression where the first-order dextral strike-slip (master) faults trend 020–030° [38]. Second-order synthetic dextral shears and antithetic sinistral displacements as well as the

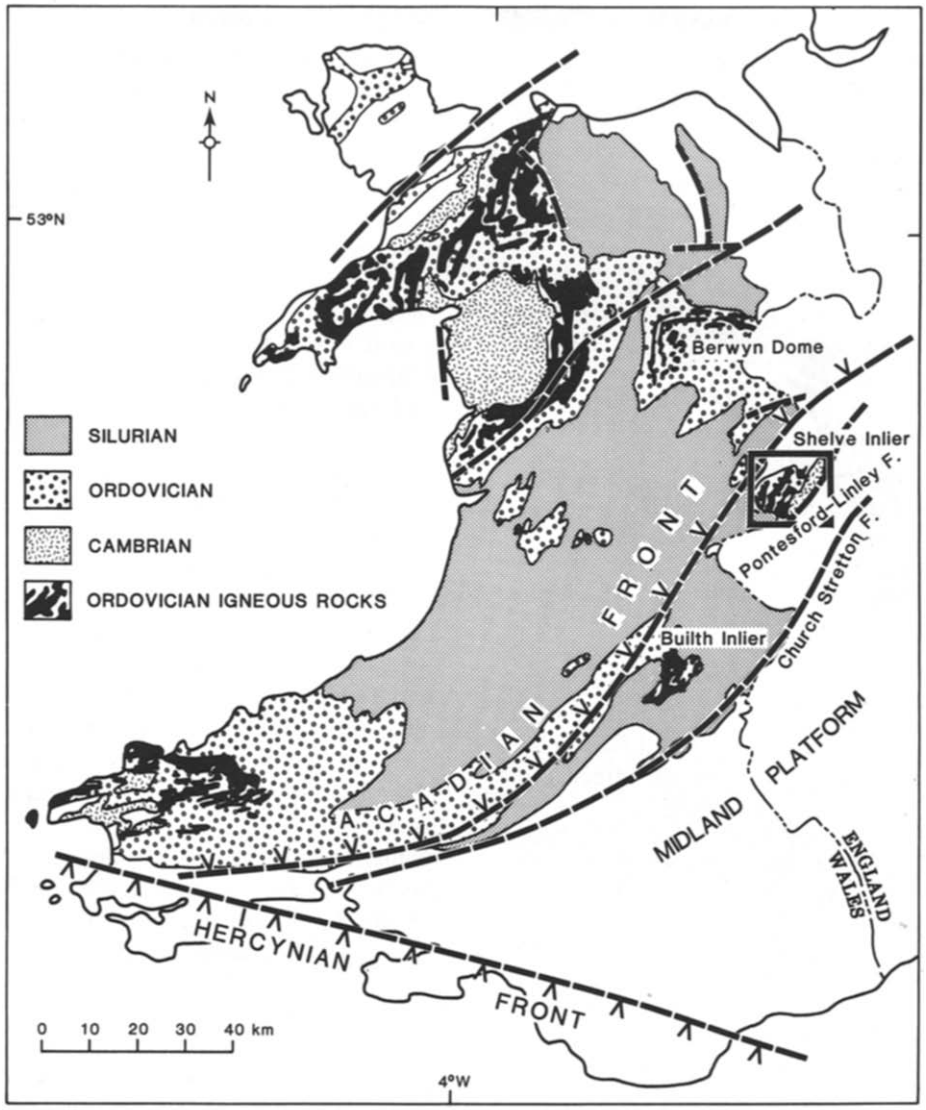



Fig. 1. Simplified geology of the Welsh Basin (after [32]) showing the location of the Shelve Inlier and other features discussed in this paper.

north–south orientation of the fold axial traces appear to be consistent with a dextral transpressional model for the deformation [38].

The Shelve Inlier exposes an Ordovician sequence of Arenig to Caradoc age. Apart from minor Caradoc tuffs, the extrusive rocks in the Shelve Inlier (Fig. 2) are within the Stapeley Volcanic Formation of Llanvirn age, which comprises a Lower Tuff and Shale Member, and overlying Todleth and Lan Fawr Members [40]. The volcanic units contain basaltic to rhyolitic lavas and pyroclastics. The majority of the intrusive rocks exposed in the Shelve Inlier are doleritic sills, dikes, and larger intrusive bodies including the ones at Corndon Hill and Snead (Fig. 2). Sedimentation within the Stapeley Volcanic Formation appears to have been influenced by magmatic doming associated with the Corndon Hill Intrusion (Fig. 2), and there is geochemical evidence for a co-magmatic relationship between the lavas and the main dolerite suite [41]. However, the dolerites can be seen to intrude Caradoc sediments, indicating a post-middle Ordovician age for at least some of the intrusives. Although the

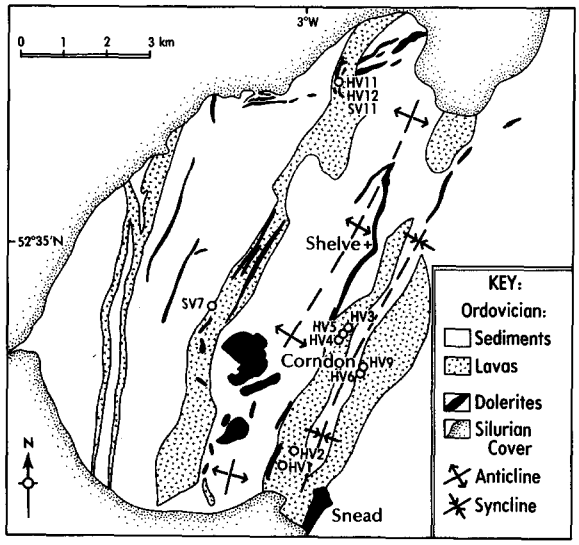



Fig. 2. Simplified geologic map of Shelve Inlier (after [28]) showing the locations of sampling sites within the Stapeley Volcanic Formation.

volcanics are characterized by fresh clinopyroxene [41], there is evidence of metamorphism in the prehnite-pumpellyite facies [42]. Reset Rb/Sr ages suggest that a chemical disturbance affected the middle Ordovician igneous rocks in the late Ordovician, perhaps associated with the culmination of volcanism in Caradoc time [41].

As part of a paleomagnetic reconnaissance of the Ordovician igneous rocks of southern Britain, we collected samples from both intrusive and extrusive units whithin the Shelve Inlier during the 1987 field season. Preliminary paleomagnetic results from the volcanic units were encouraging, so the following year we returned to the Shelve area and collected 10 or more oriented samples from each of 13 sites within the Lower Tuff and Shale Member of the Stapeley Volcanic Formation (Fig. 2). Samples were drilled in the field with a portable gasoline-powered rock drill, and oriented with a Brunton compass and clinometer. Care was taken to distribute the sites between easterly, and westerly dipping fold limbs (Fig. 2) for purposes of a paleomagnetic fold test.

## 3. Laboratory methods

Samples were cut into standard $2.2 \times 2.5$ cm specimens. During the reconnaissance study, we had employed both thermal and alternating field demagnetization techniques. The thermal method produced superior results, so all samples from the second collection were subjected to thermal demagnetization using 12 or more demagnetization steps. Samples were heated in a three-zone furnace powered by a Lindberg furnace controller, and protected from the Earth's magnetic field by three cylindrical mu-metal shields. The residual field is less than 50 nT in the furnace and less than 5 nT in the cooling chamber. The remanent magnetization of specimens was measured with the three-axis superconducting magnetometer at Louisiana State University. Demagnetization data were plotted on orthogonal demagnetization diagrams [43] for inspection, and the directions of the component magnetizations were determined from linear demagnetization trends using principal component analysis [44].

In order to constrain the mineralogy of the magnetic carriers, representative samples were subjected to the stepwise acquisition of isothermal remanent magnetization (IRM) using an impulse magnetizer with a maximum peak field of 1.1 T. Following the IRM acquisition experiments, samples were given a composite IRM (1.1 T in one direction, and 0.2 T in the antiparallel direction) which was then thermally demagnetized [45]. This method allows the discrimination of hematite from goethite among high-coercivity carriers, and magnetite from maghemite or monoclinic pyrrhotite among low-coercivity carriers.

## 4. Results

The representative demagnetization diagrams shown in Fig. 3 indicate typical behavior of Stapeley Volcanic Formation samples during thermal demagnetization. Nearly all of the samples that demagnetized stably were found to contain a magnetization with easterly to southeasterly declinations and steep down inclinations after correction for bedding tilt. This magnetization direction is the characteristic magnetization in our samples, and it generally has distributed unblocking temperatures up to 500° or 525°C. By this temperature, 90% or more of the original NRM is usually removed. However, after treatment at higher temperatures the magnetization looses its directional stability. This loss of stability is accompanied by increases of the low-field susceptibility and sometimes by marked increases in remanent intensity,

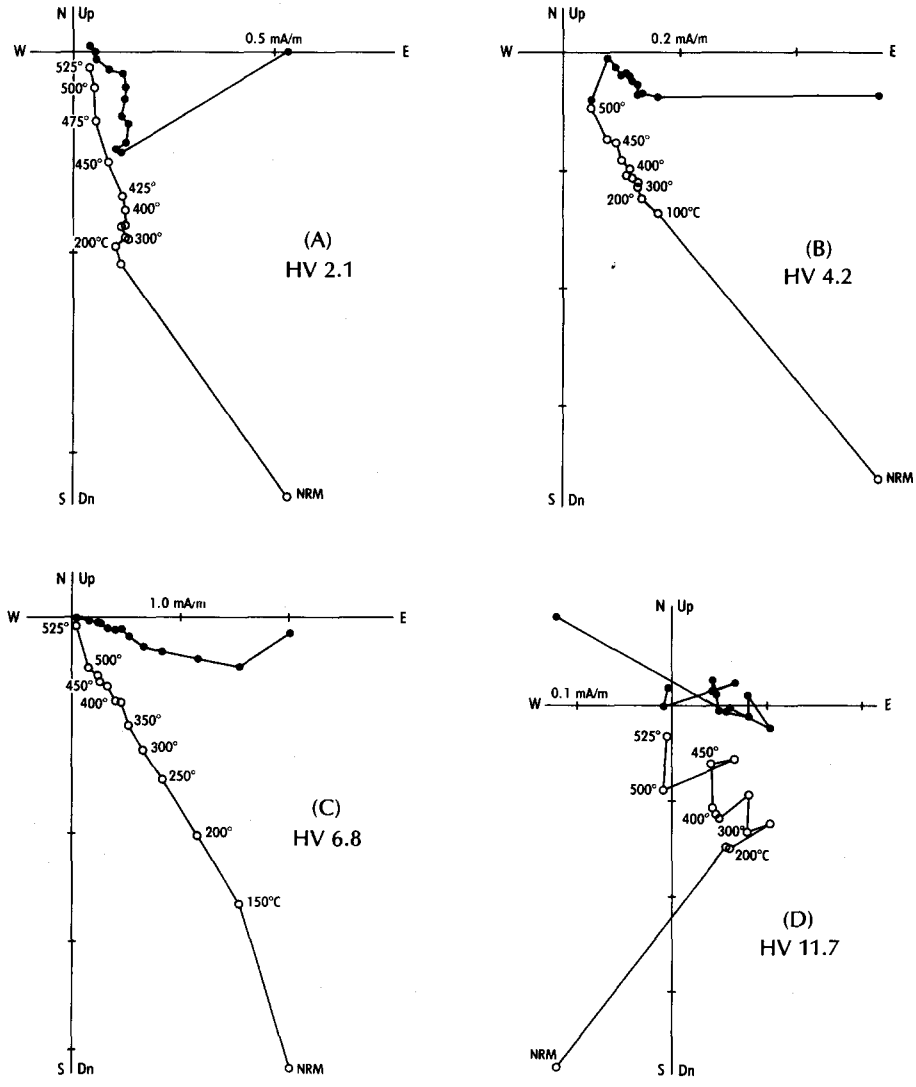



Fig. 3. Orthogonal demagnetization diagrams [43] showing the behavior of Stapeley Volcanic Formation samples during stepwise thermal demagnetization. A and B are for samples taken from easterly-dipping fold limbs, and C and D are from beds with westerly dips. Coordinates are corrected for structural tilt. Open circles are projections of the magnetization vectors on the east-west vertical plane and solid circles are projections onto the horizontal plane. Data from treatments above 500° or 525°C are not shown for clarity (see text).

reflecting the creation of new magnetic phases during thermal treatment. As a result of this behavior, no magnetization components with blocking temperatures above about 525°C were well-resolved in our data.

Fig. 3A shows an example of a multicomponent magnetization, commonly observed in our data. After removal of a viscous(?) magnetization, a component with a southerly and shallow direction in field coordinates is revealed which unblocks between about 300° and 425°C. This magnetization is usually poorly defined due to the apparent overlap of its blocking temperature spectrum with that of the characteristic magnetization. Unfortunately, it was not well-resolved in enough samples for a detailed analysis, but we infer that it may be an overprint of late Paleozoic age, similar to the one which was found in coeval rocks at the nearby Builth Inlier [30]. The characteristic magnetization is well-defined in this specimen (Fig.

3A), and it unblocks as a single component between 450° and 525°C. Finally, a poorly-resolved magnetization with unblocking temperatures above 525°C is indicated by the fact that the characteristic magnetization does not decay towards the origin of the diagram, but rather towards a northerly and down direction in field coordinates. This higher unblocking-temperature magnetization may reside in hematite, and be the result of recent weathering. Fig. 3B and C show data in which the characteristic magnetization is well-resolved, but the "late Paleozoic" magnetization found in some of the samples is not defined. Fig. 3D shows data from a sample with a relatively low intensity of natural remanence. The characteristic magnetization is clearly present in this sample, but it is not as well defined as in the other examples shown.

Samples with the weakest magnetizations before thermal treatment often did not magnetize stably. Most of the sites included some specimens that did not demagnetize with sufficient stability for the resolution of single magnetization components. We accept site data if more than half of the specimens yield resolvable magnetization components during thermal demagnetization. Unstable magnetizations were found to predominate in four of the 13 sites we collected during 1988, and these sites will not be considered further. Using the 50% acceptance criterion, two sites (SV7 and SV11) collected during the reconnaissance study, and 9 sites from the main study were judged acceptable.

The characteristic magnetization directions determined by principal component analysis for individual specimens are shown both before (Fig. 4A) and after (Fig. 4B) tilt correction. Site mean directions are indicated before (Fig. 4C) and after (Fig. 4D) correction for bedding tilt. The site data and pertinent statistical parameters for the characteristic magnetization are listed in Table 1. The fold test is clearly positive, and is significant at the 99% confidence level [47], indicating that the characteristic magnetization was acquired prior to the Ashgill folding.

Acquisition of isothermal remanent magnetization (IRM) experiments (Fig. 5A) show that Stapeley Volcanic Formation samples reach saturation, or near saturation at magnetizing fields of about 200 mT, indicating the dominance of low coercivity magnetic phases in the samples. Thermal demagnetization of a composite IRM (Fig. 5B) shows that low coercivity phases have blocking temperatures up to 550° or 575°C, indicating that magnetite is present, and probably dominates the population of magnetizable grains in the sam-

TABLE 1

Summary of site paleomagnetic data for the characteristic magnetization of Shelve Inlier volcanic samples

| Site i.d. | $N/N_0$ | Mean MAD | In situ $D/I$ (°) | Bedding $D/I$ (°) | $k$ | $\alpha_{95}$ (°) |
|---|---|---|---|---|---|---|
| HV1 | 6/12 | 8.1 | 259.8/+72.6 | 135.6/+68.2 | 52.9 | 9.3 |
| HV2 | 7/8 | 4.4 | 271.3/+72.1 | 137.0/+69.1 | 195.1 | 4.3 |
| HV3 | 8/11 | 2.6 | 287.1/+58.1 | 124.7/+57.3 | 80.4 | 6.8 |
| HV4 | 11/15 | 7.5 | 295.7/+64.4 | 114.4/+58.5 | 44.2 | 6.9 |
| HV5 | 9/10 | 12.0 | 304.6/+53.6 | 101.2/+68.5 | 63.1 | 6.1 |
| HV6 | 14/14 | 4.5 | 119.6/−10.6 | 123.7/+59.1 | 152.7 | 3.2 |
| HV9 | 6/12 | 8.8 | 119.6/+5.2 | 129.7/+74.7 | 30.8 | 12.3 |
| HV11 | 9/13 | 7.0 | 105.9/+11.7 | 98.9/+68.4 | 51.0 | 7.3 |
| HV12 | 6/11 | 3.8 | 108.0/+20.0 | 98.9/+76.4 | 35.1 | 11.5 |
| SV7 | 10/15 | 12.8 | 118.4/+1.4 | 111.2/+68.0 | 26.0 | 9.7 |
| SV11 | 9/12 | 4.2 | 104.3/+17.0 | 92.8/+72.3 | 172.4 | 3.9 |
| Means: | | | 115.5/+54.4<br>$k = 2.2, \alpha_{95} = 41.2°$ | 116.4/+67.9<br>$k = 89.4$, $\alpha_{95} = 4.9°$ | | |
| Pole | | | | 26.6°N, 36.1°E<br>$\delta p = 6.9°$, $\delta m = 8.2°$ | | |

$N/N_0$ is the number of specimen directions used compared with the number of specimens thermally demagnetized; Mean MAD is the average maximum angular deviation for the site [44], $D$ is declination, and $I$ is inclination. $k$, $\delta p$, $\delta m$, and $\alpha_{95}$ are statistical parameters [46].

ples. Some of the samples also contain small quantities of high-coercivity minerals that have blocking temperatures up to 675°C, indicating that hematite is also present. The presence of hematite in some of the samples is consistent with our field observation that some samples appear to be slightly weathered, particularly those taken close to visible joints. Minor hematite is also consistent with the observation that the characteristic magnetization component does not always decay towards the origin of orthogonal demagnetization diagrams (e.g. Fig. 3A, 3B), but rather towards a recent field direction.

## 5. Discussion

The characteristic magnetization isolated in this study was clearly acquired prior to folding in the Shelve area. Since the folding in the region is an Ashgill event, the characteristic magnetization was acquired during either the middle or late Ordovician, an extremely tight constraint for a magnetization of early Paleozoic age. However, there are reasons to suspect that the characteristic magnetization may not have been acquired as a primary thermoremanence, but rather represents a complete chemical or thermochemical remagnetization later in the Ordovician, but prior to folding. The single (apparently reversed) polarity and distributed unblocking temperatures of the characteristic remanence suggest that remagnetization may have occurred. Moreover, our own thin section observations of representative samples indicate that these volcanic rocks have suffered pervasive alteration. In addition to primary igneous minerals, we observed abundant secondary phases including chlorite, zeolites, sphene (after titanomagnetite), framboidal pyrite, and silicification. Our observations are consistent with the report that these and other

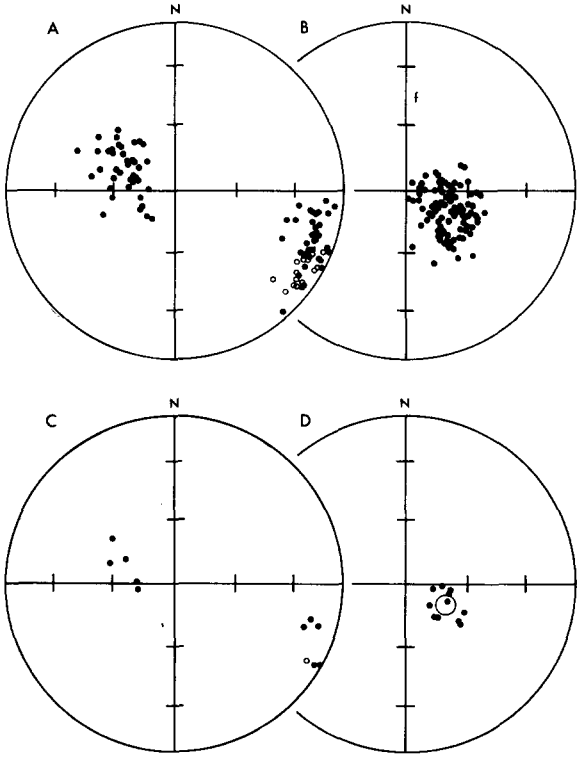



Fig. 4. Characteristic paleomagnetic directions in equal area projection. Solid circles are on the lower-hemisphere and open circles are on the upper one. A, B. Specimen directions before (A) and after structural correction (B). C, D. Site mean directions before (C) and after structural correction (D). The large circle in (D) indicates the circle of 95% confidence about the mean direction; the mean itself is not shown for clarity.

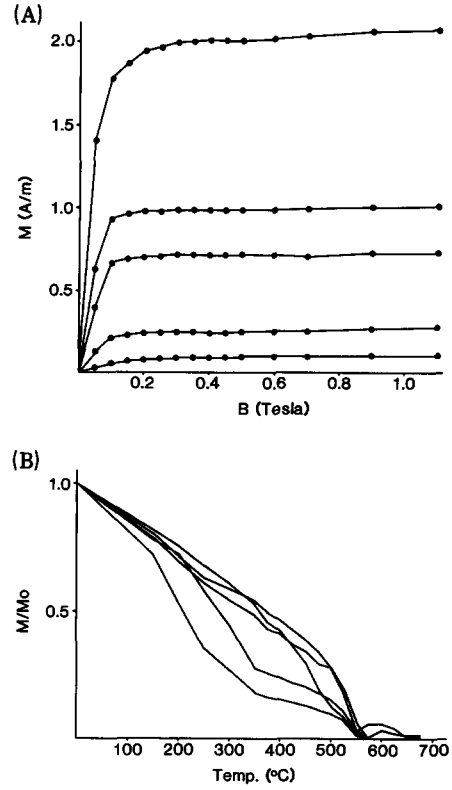



Fig. 5. (A) Results of acquisition of IRM experiments. (B). Thermal demagnetization of IRM for the same 5 specimens from the Stapeley Volcanic Formation (see text).

Welsh Ordovician igneous rocks have undergone prehnite–pumpellyite metamorphism [42]. Given the highly altered state of the samples, particularly with respect to the primary remanence-carrying oxides, preservation of the original thermoremanent magnetization is highly questionable. Pending further petrologic and geochemical investigation of these rocks, we suggest that the characteristic magnetization discussed in this paper is likely to have been acquired in late Ordovician time, and that its origin is associated with the thermochemical disturbance of that age [41].

Our mean inclination of 68° indicates that southern Britain occupied a latitude of about 51° during late Ordovician time. Due to the similarity of the Ordovician shelf faunas of southern Britain and Gondwana [22], we infer that southern Britain was in the southern hemisphere, at 51° south. Since middle to late Ordovician paleolatitudes for Gondwana and Laurentia are well-determined on the basis of both paleoclimate indicators and paleomagnetic data [1], our result can be used to place constraints on the paleogeographic setting of southern Britain in a middle to late Ordovician global context (Fig. 6). Southern Britain must have been very close to the northern margin of Africa or South America at this time. Time is consistent with the inference that Southern Britain was a "Peri-Gondwanide" terrane during the early Paleozoic, along with Hercynian Europe and Avalon [1]. However, since individual paleomagnetic poles give no paleolongitude information, we cannot determine if southern Britain was attached to Gondwana at this time, or was a nearby microcontinent. It is nonetheless quite clear that a wide late Ordovician Iapetus Ocean separated Laurentia on the one hand, and Gondwana and the Peri-Gondwanide terranes, including southern Britain, on the other.

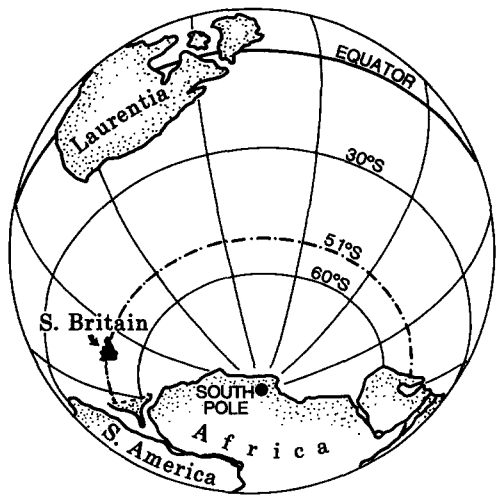



Fig. 6. Map of middle to late Ordovician paleopositions for Laurentia and Gondwana based on paleomagnetic data (after [1]). Note that paleolatitudes are constrained but paleolongitudes are not. Southern Britain is positioned according to the result reported in this paper. The dashed line at 51°S indicates the range of permissible locations for southern Britain in this reconstruction.

Paleomagnetic and paleoclimatologic evidence suggests that Gondwana moved northward during the Silurian [10,11], indicating that the northern margin of Africa/South America could have moved into a latitude sufficiently low for collision with Laurentia by Acadian time [1]. We infer that southern Britain moved northward with Gondwana during the Silurian, and that final closure of Iapetus in Britain is marked by the widespread Acadian deformation of early Devonian age (see [33,34]). This inference is consistent with the observed endemicity of freshwater faunas on the opposing margins of Iapetus in Britain until early Devonian time [48].

## 6. Conclusions

Paleomagnetic results similar to the easterly and steeply down direction found in the present study have been reported previously from several Ordovician intrusive units within the Shelve Inlier [28] and from the Ordovician Moel-y-Golfa Andesite of the nearby Breidden Hill Inlier [29]. However, these results have previously been considered either "anomalous" (e.g. [12]) or magnetized in a post-Ordovician field [29]. The new results from the Shelf Inlier indicate that this magnetization direction must be viewed as truly representative of the late Ordovician paleomagnetic field in southern Britain.

The difference between the result of this investigation and that of the recent study of Llanvirn volcanic rocks of the Builth Inlier [30] is not explained at present, but some comment on this discrepancy appears in order. The Builth magnetization direction of $D = 182°$, $I = 55°$ (after tilt correction) is flattened by 13° and rotated 65°

(clockwise) with respect to the new Shelve result. The difference in declination might be attributed to tectonic rotations associated with post-middle Ordovician strike-slip faulting that affected the region [39]. Samples for the Builth Inlier study [30] were obtained from the Llanelwedd Quarry which is within a structural domain that is postulated to have been rotated clockwise with respect to adjacent domains [39], and could therefore be rotated with respect to the Shelve Inlier rocks. However, the discrepancy in inclination between the two studies is significant and cannot be attributed to tectonic rotation, so some other explanation must be sought. One possibility is that the Builth and Shelve magnetizations are not the same age. Additional detailed study of the Builth Inlier rocks with particular attention to constraining the magnetization age may help to resolve this problem.

The new paleomagnetic data from the Shelve Inlier provide strong evidence that southern Britain occupied a high southerly paleolatitude during the late Ordovician, and could have been attached to the northern margin of West Gondwana together with Hercynian Europe and Avalon. Within the context of a global reconstruction for the middle/late Ordovician [1], it is apparent that the Iapetus Ocean in Britain had a latitudinal width of some 30° at this time (Fig. 6). The existence of a wide Iapetus in Britain during the later Ordovician argues against the proposal of final closure of the ocean during the Ordovician, which has been suggested by a reevaluation of sedimentary facies within the Southern Uplands of Scotland (e.g. [16,17]). We regard it much more likely that the ocean closed gradually during the Silurian, with final docking of southern Britain against Laurentia associated with the Acadian Orogeny in early Devonian time.

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